

Computer Networking Homework #4

- (a) How many bytes are there in the header field of a typical IPv4 datagram without optional header field?
(b) What is the purpose of the time-to-live (TTL) field in the header of a IPv4 datagram? How does this field function when a IPv4 datagram is processed by a router?

(a) 20 bytes

(b) TTL field will decrease when passing every router, to avoid packet stay in network forever

- Consider a router that interconnects three subnets: Subnet 1, Subnet 2, and Subnet 3. Suppose all of the interfaces in each of these three subnets are required to have the prefix 223.3.16.0/24. Also suppose that Subnet 1 is required to support up to 120 interfaces, and Subnets 2 and 3 are each required to support up to 50 interfaces. Provide three network addresses (of the form a.b.c.d/x) that satisfy these constraints.

Subnet 1: 223.3.16.0/25

Subnet 2: 223.3.16.128/26

Subnet 3: 223.3.16.192/26

1 1
7 6 5 4 3 2 1 0
128 64 32 16 8 4 2 1

- What is a private network address? Should a datagram with a private network address ever be present in the larger public Internet, why or why not?

(1) ~~A private network address is obtain by the local router.~~

This address is non-routable in the public network, only used in local network

(2) This kind of address won't ever be present in the public Internet because the NAT function in router will change it to the IP from the ISP

- (a) Explain the **tunneling** technique used for the IPv4-to-IPv6 transition. (b) Do you agree with the following statement: when IPv6 tunnels through IPv4 router, IPv6 treats the IPv4 tunnels a link-layer protocols?

(a) Tunneling will make a whole IPv6 packet a payload of a IPv4 packet

(b) ~~No, because this kind of modify only change the data managed in network layer, it has nothing to do with the link layer.~~

Yes, because for the IPv4 routers, they provide a method to route IPv6 packets, which makes the tunnel like a link layer for IPv6

5. What is the meaning of “match plus action” operation of a router or switch? In the case of destination-based forwarding packet switch, what is matched and what is the action taken? In the case of an SDN, name 2 fields that can be matched and 2 actions that can be taken.

(a) Match plus action is a policy for treating the incoming packets, router can match the characteristics of the packet and take the corresponding action to it

(b) It matches the destination IP and does the forwarding action

(c) Matched field: Source IP, Source Port / Actions: Forward, Drop